



**RAJALAKSHMI
ENGINEERING COLLEGE**

An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

**DEPARTMENT OF COMPUTER SCIENCE AND
ENGINEERING**

CS23432 – Software Construction

(REGULATION 2023)

**RAJALAKSHMI ENGINEERING COLLEGE
Thandalam, Chennai-602015**

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Year / Branch / Section: II/CSE/FE

Semester: IV

Academic Year: 2025-2026

INDEX

S.No.	Date	Title
1.		Azure Devops Environment Setup.
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3.		Setting Up Epics, Features, And User Stories for Project Planning.
4.		Sprint Planning.
5.		Poker Estimation.
6.		Designing Class and Sequence Diagrams for Project Architecture.
7.		Designing Architectural and ER Diagrams for Project Structure.
8.		Testing – Test Plans and Test Cases.
9.		Load Testing and Pipelines.
10.		GitHub: Project Structure & Naming Conventions.

EXP NO: 1	AZURE DEVOPS ENVIRONMENT SETUP
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Aim:

To set up and access the Azure DevOps environment by creating an organization through the Azure portal.

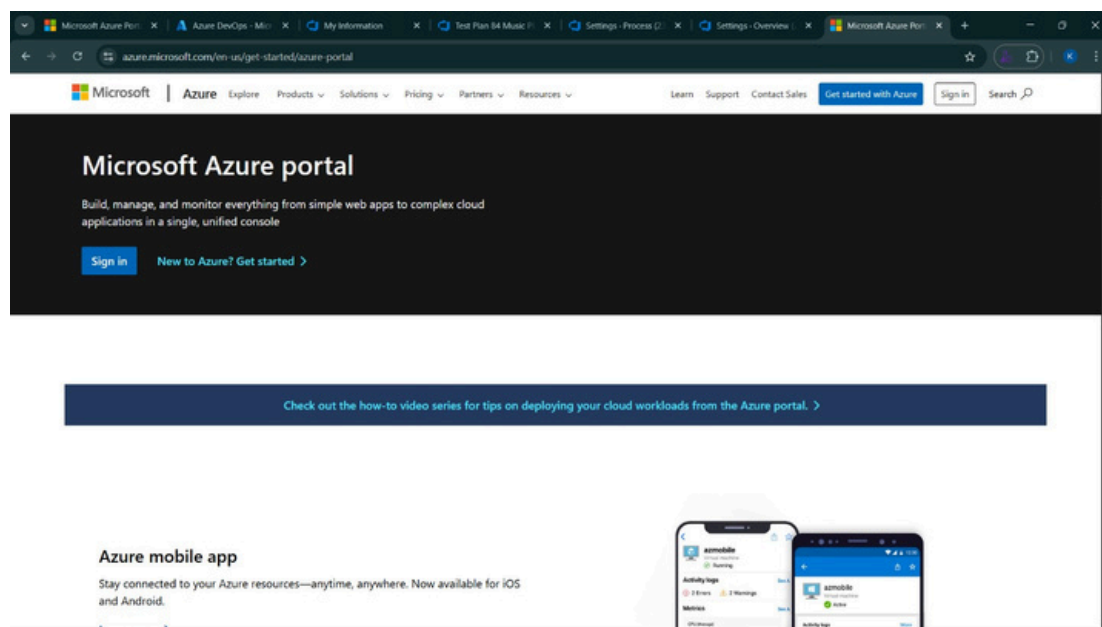
INSTALLATION

1. Open your web browser and go to the Azure website:

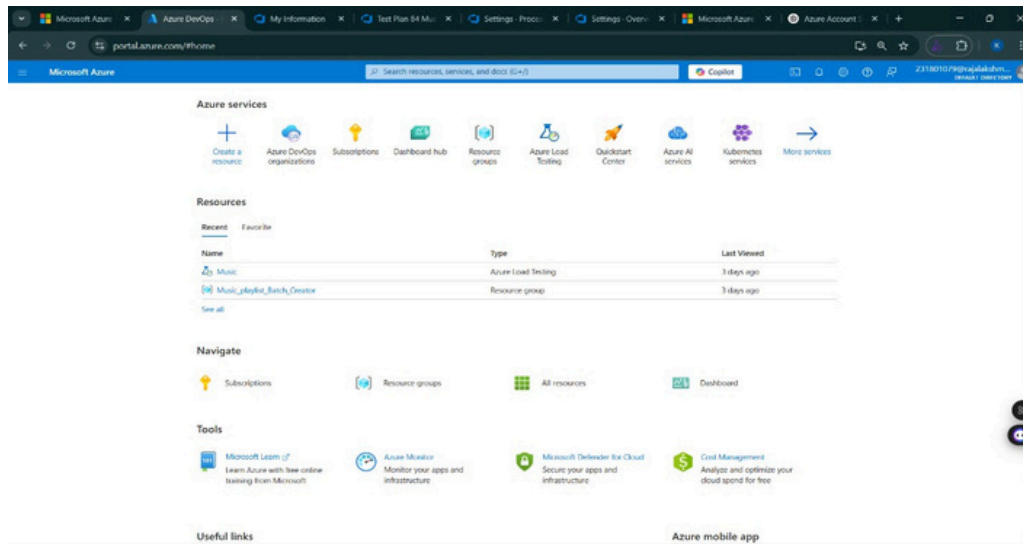
<https://azure.microsoft.com/en-us/get-started/azure-portal>.

Sign in using your Microsoft account credentials.

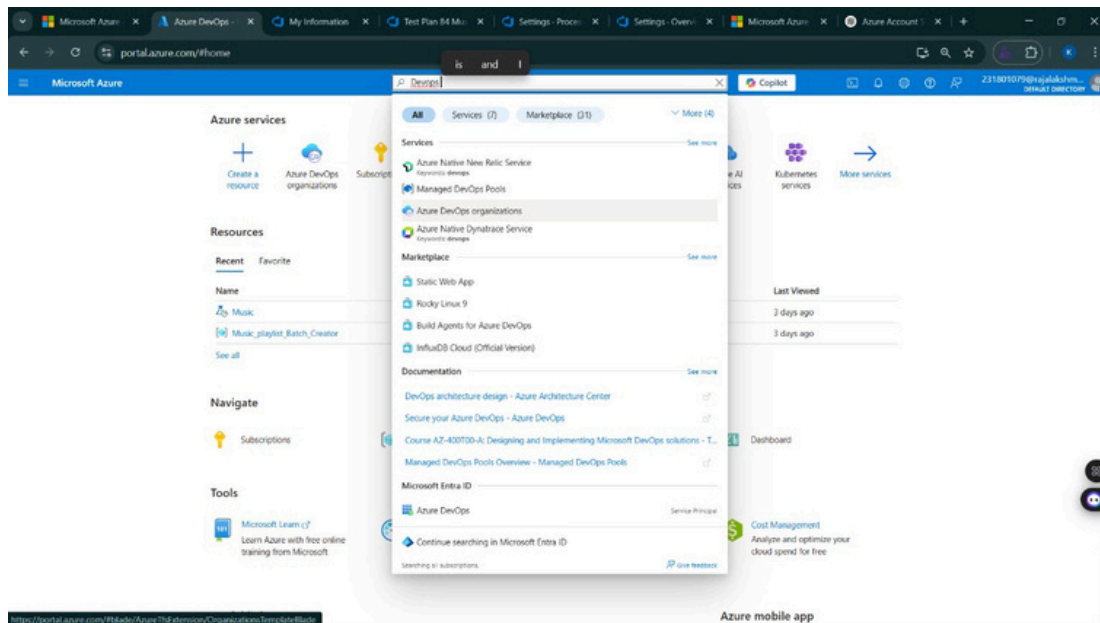
If you don't have a Microsoft account, you can create one here: <https://signup.live.com/?lic=1>



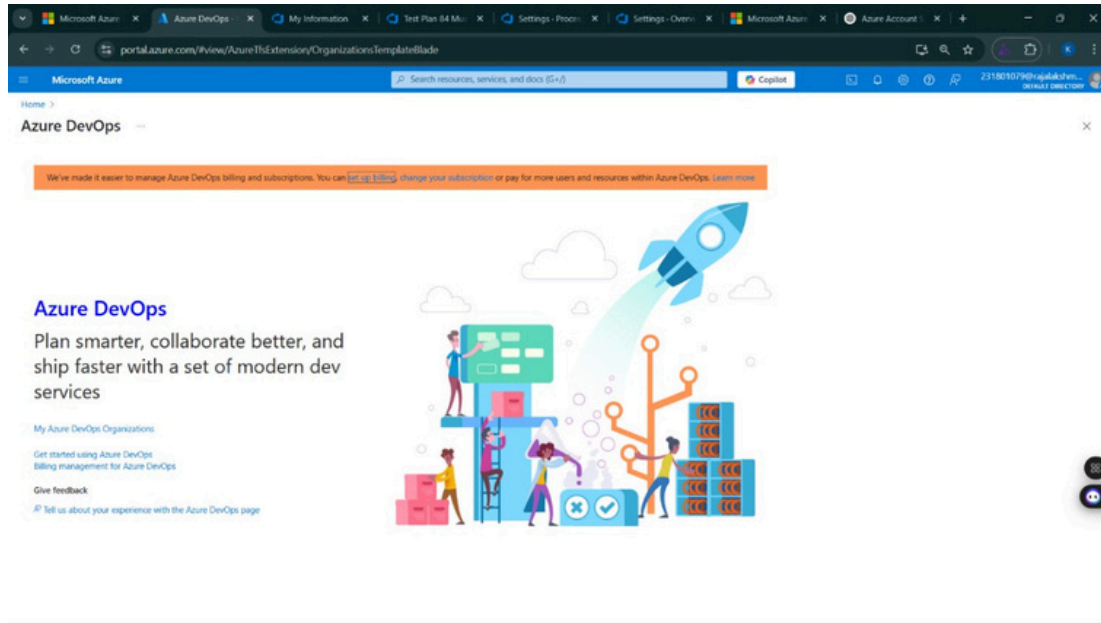
2. Azure home page



3. Open DevOps environment in the Azure platform by typing **Azure DevOps Organizations** in the search bar.



4. Click on the **My Azure DevOps Organization** link and create an organization and you should be taken to the Azure DevOps Organization Home page.



Result:

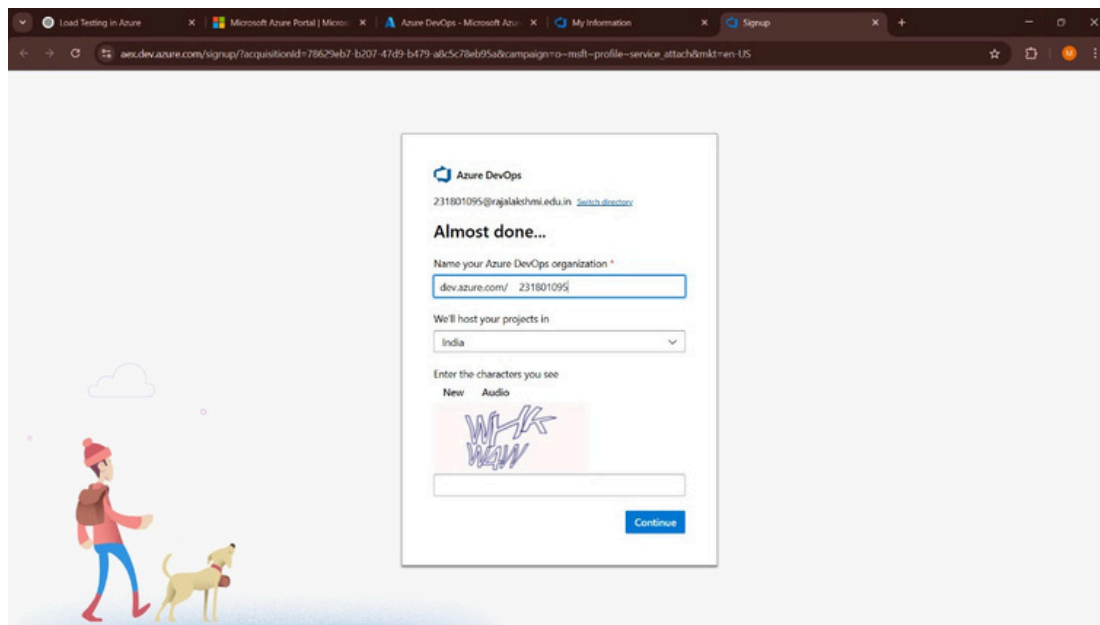
Successfully accessed the Azure DevOps environment and created a new organization through the Azure portal.

EXP NO: 2	AZURE DEVOPS PROJECT SETUP AND USER STORY MANAGEMENT
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Aim:

To set up an Azure DevOps project for efficient collaboration and agile work management.

1.Create An Azure Account



2.Create the First Project in Your Organization

- After the organization is set up, you'll need to create your first **project**. This is where you'll begin to manage code, pipelines, work items, and more.
- On the organization's **Home page**, click on the **New Project** button.
- Enter the project name, description, and visibility options:
 - Name:** Choose a name for the project (e.g., **LMS**).
 - Description:** Optionally, add a description to provide more context about the project.
 - Visibility:** Choose whether you want the project to be **Private** (accessible only to those invited) or **Public** (accessible to anyone).
- Once you've filled out the details, click **Create** to set up your first project.

Create new project

×

Project name *

PRODUCT REVIEW AND RATING SYSTEM SC-PROJECT

Description

The Intelligent Multi-Criteria Product Review and Rating System is a web application where users can review products and rate them based on different criteria such as quality, price, and usability. The system also analyzes review text using sentiment analysis to determine whether the feedback is positive, negative, or neutral.

Visibility

Private

Only people you give access to will be able to view this. Want to create a public project? [Try GitHub](#)

^ Advanced

Version control ⓘ

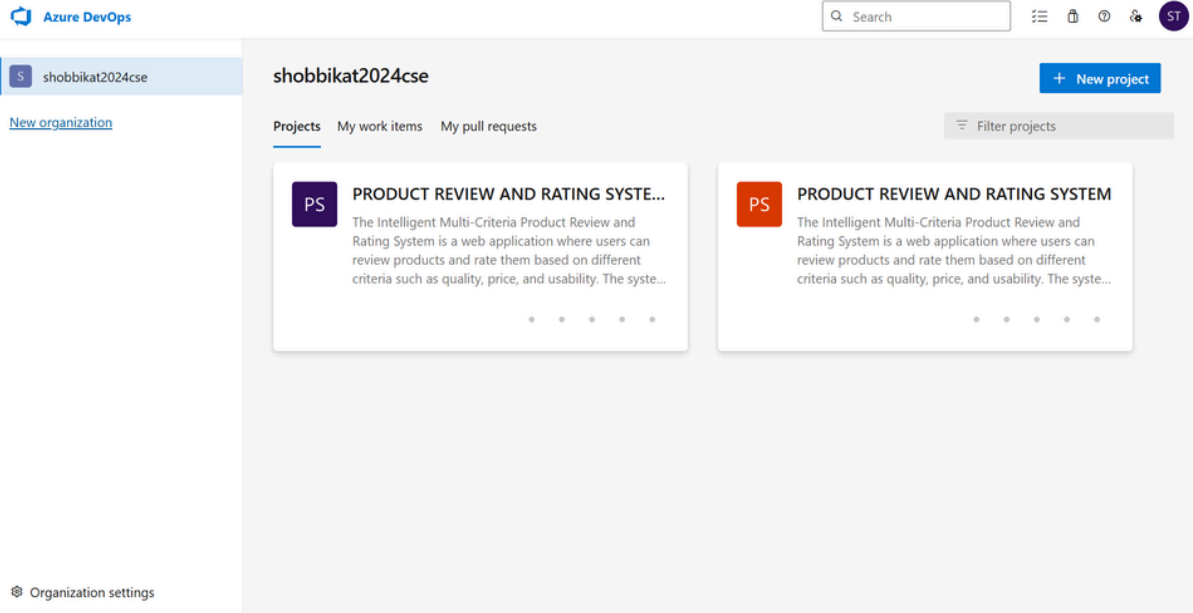
Git

Work item process ⓘ

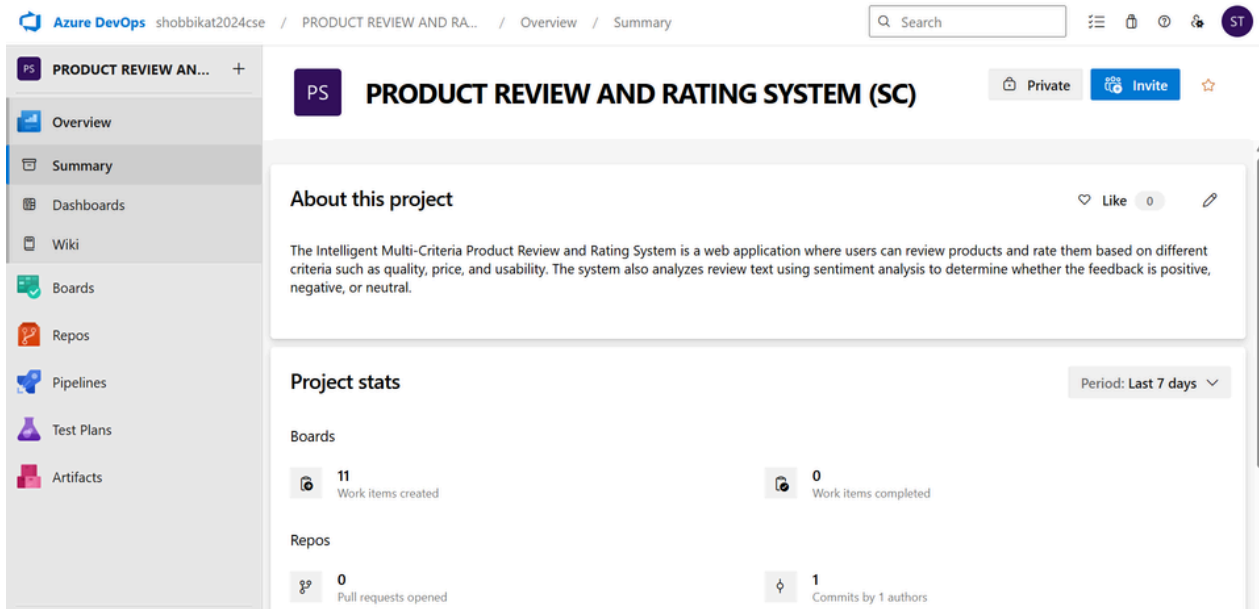
Agile

Cancel>Create

3.Once logged in, ensure you are in the correct organization. If you're part of multiple organizations, you can switch between them from the top left corner (next to your user profile). Click on the Organization name, and you should be taken to the Azure DevOps Organization Home page.



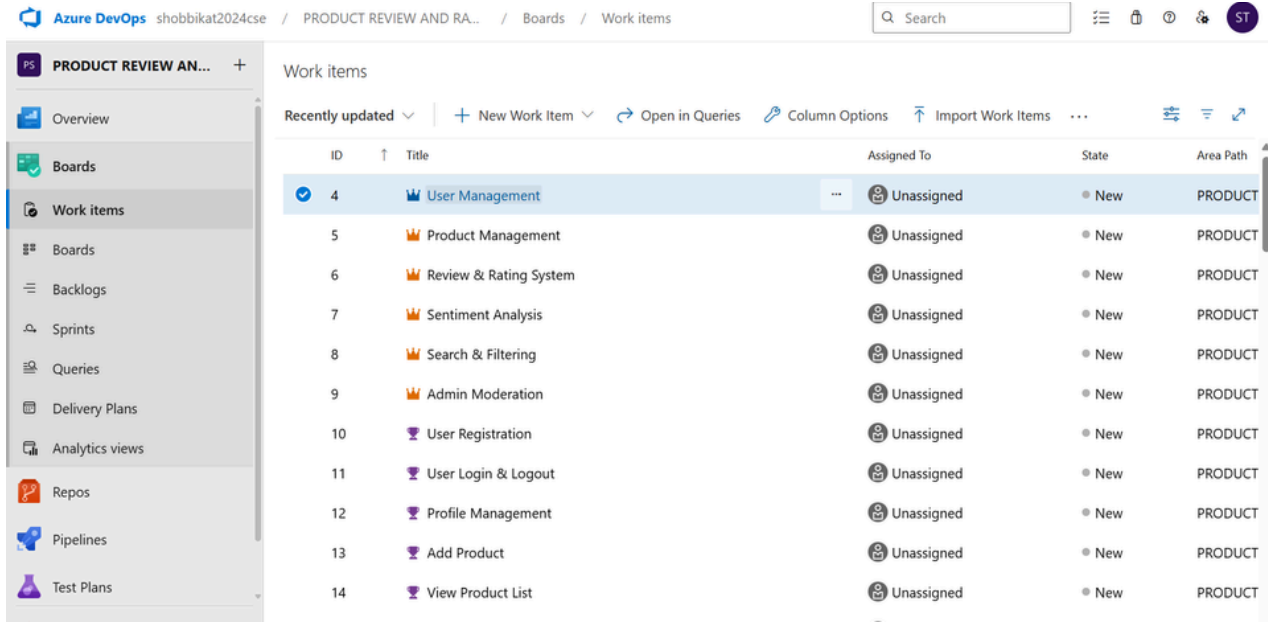
4.Project dashboard



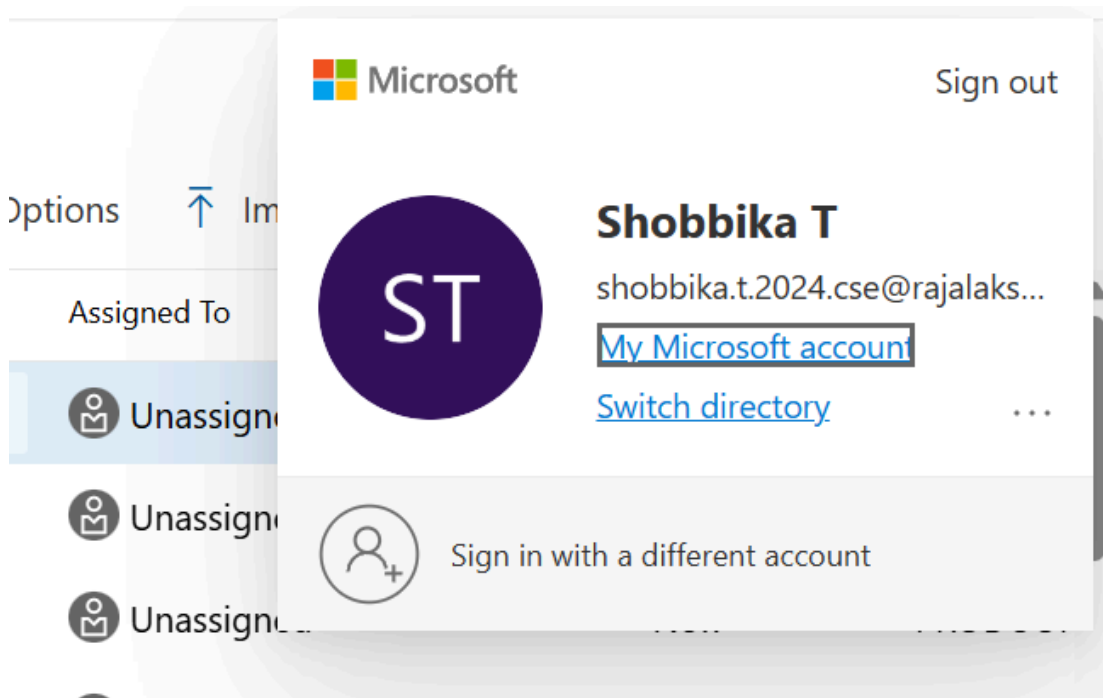
5. To manage user stories:

a. From the **left-hand navigation menu**, click on **Boards**. This will take you to the main **Boards** page, where you can manage work items, backlogs, and sprints.

b. On the **work items** page, you'll see the option to **Add a work item** at the top. Alternatively, you can find a + button or **Add New Work Item** depending on the view you're in. From the **Add work item** dropdown, select **User Story**. This will open a form to enter details for the new User Story.



ID	Title	Assigned To	State	Area Path
4	User Management	Unassigned	New	PRODUCT
5	Product Management	Unassigned	New	PRODUCT
6	Review & Rating System	Unassigned	New	PRODUCT
7	Sentiment Analysis	Unassigned	New	PRODUCT
8	Search & Filtering	Unassigned	New	PRODUCT
9	Admin Moderation	Unassigned	New	PRODUCT
10	User Registration	Unassigned	New	PRODUCT
11	User Login & Logout	Unassigned	New	PRODUCT
12	Profile Management	Unassigned	New	PRODUCT
13	Add Product	Unassigned	New	PRODUCT
14	View Product List	Unassigned	New	PRODUCT



Microsoft Sign out

Shobbika T
shobbika.t.2024.cse@rajalaks...
[My Microsoft account](#)
[Switch directory](#)

Sign in with a different account

Result:

Successfully created an Azure DevOps project with user story management and agile workflow setup.

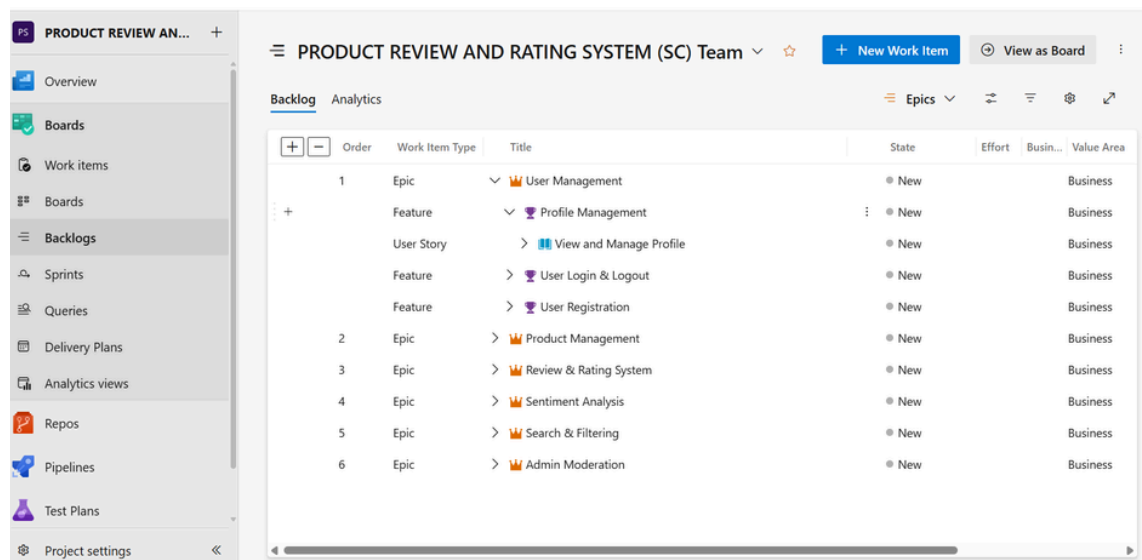
EXP NO: 3

SETTING UP EPICS, FEATURES, AND USER STORIES FOR PROJECT PLANNING

Aim:

To learn about how to create epics, user story, features, backlogs for your assigned project.

Create Epic, Features, User Stories, Task



1.Fill in Epics

USER STORY 32

32 View and Manage Profile

Pranav Shanmugam

0 Comments Add Tag

Save and Close

Follow

State

New

Area

PRODUCT REVIEW AND RATING SYSTEM (SC)

Reason

New

Iteration

PRODUCT REVIEW AND RATING SYSTEM (SC)\Iteration 1

Updated by Shobbika T: Apr 9

Details

2

0

Description

Acceptance Criteria

- User should be able to view profile details.
- User should be able to edit profile information.
- Changes should be saved successfully.
- Updated information should appear immediately.

Discussion

Planning

Story Points

3

Priority

2

Risk

Classification

Value area

Business

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your

Result:

Thus, the creation of epics, features, user story and task has been created successfully.

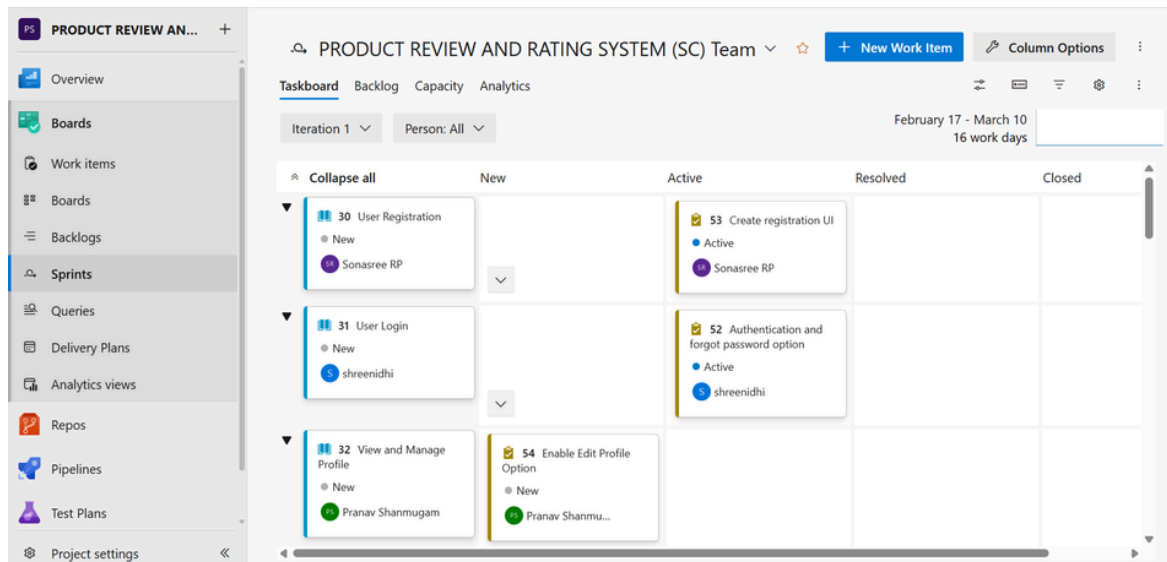
EXP NO: 4

SPRINT PLANNING

Aim:

To assign user story to specific sprint for the Product review and rating system Project.

SprintPlanning Sprint1



Sprint 2

PS

PRODUCT REVIEW AN... +

Overview

Boards

Work items

Boards

Backlogs

Sprints

Queries

Delivery Plans

Analytics views

Repos

Pipelines

Test Plans

PRODUCT REVIEW AND RATING SYSTEM (SC) Team

+ New Work Item

Column Options

Taskboard

Backlog

Capacity

Analytics

Iteration 1

Person: All

February 17 - March 10
16 work days

Collapse all

New

Active

Resolved

Closed

33 Add New Product

New

Shobbika T

34 Delete Product

New

shreenidhi

35 Edit Product Details

New

Sonasree RP

Sprint 3

PS

PRODUCT REVIEW AN... +

Overview

Boards

Work items

Boards

Backlogs

Sprints

Queries

Delivery Plans

Analytics views

Repos

Pipelines

Test Plans

Project settings

PRODUCT REVIEW AND RATING SYSTEM (SC) Team

+ New Work Item

Column Options

Taskboard

Backlog

Capacity

Analytics

Iteration 3

Person: All

April 13 - April 17
5 work days

Collapse all

New

Active

Resolved

Closed

38 Edit or Delete Review

New

Sonasree RP

40 Rate Product

New

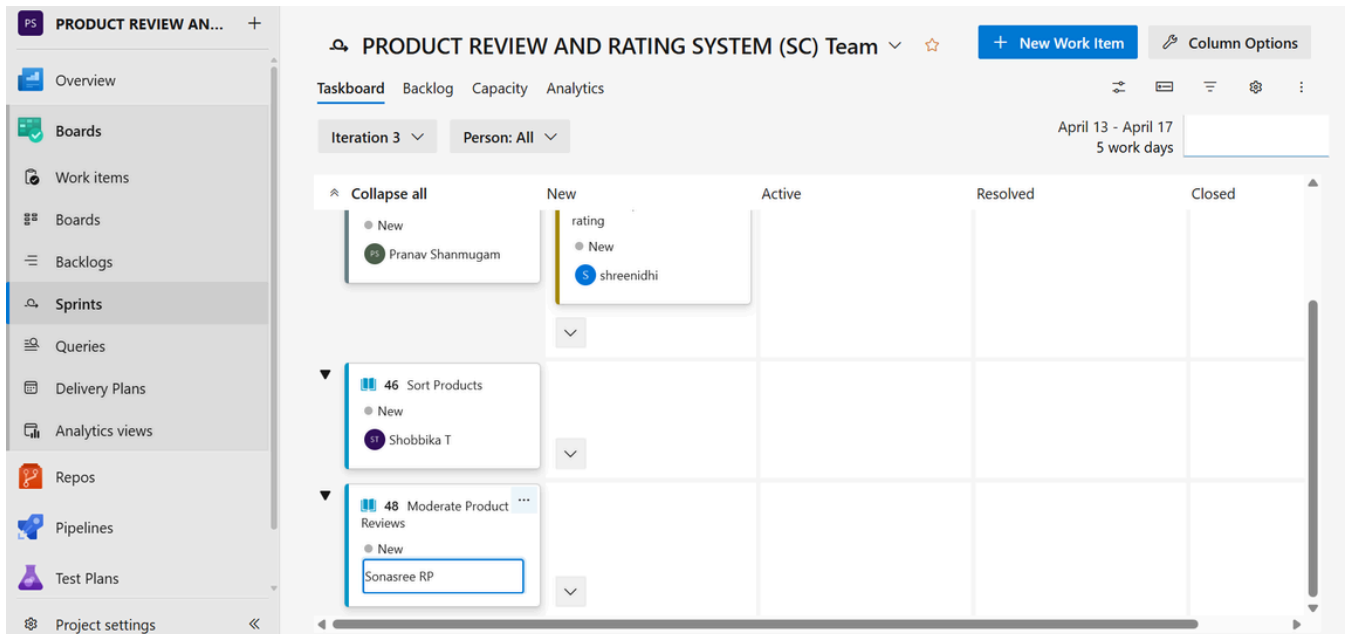
Pranav Shanmugam

70 Implement star UI for rating

New

shreenidhi

Sprint 4



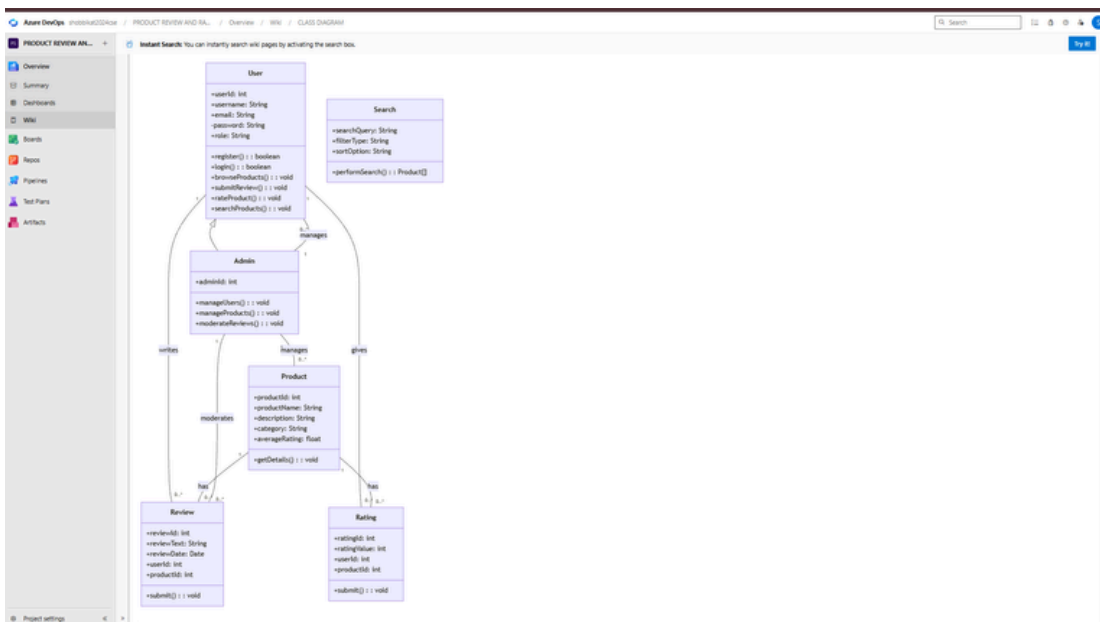
Result:

The Sprints are created for the Product Review and Rating System Project.

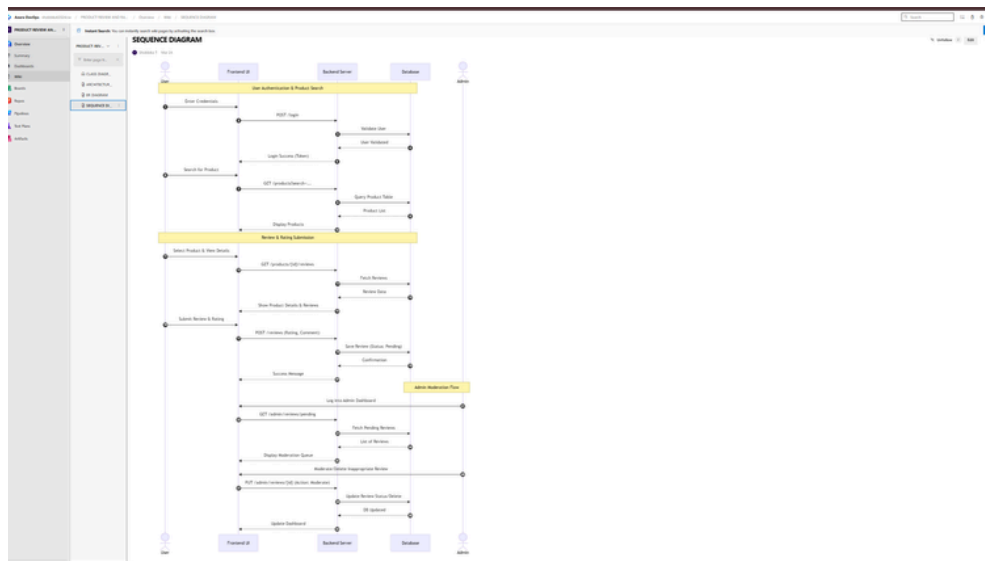
Aim:

To Design a Class Diagram and Sequence Diagram for the given Project.

6A. Class Diagram



6B. Sequence Diagram



Result:

The Class Diagram and Sequence Diagram is designed Successfully for the Product Review and Rating System.

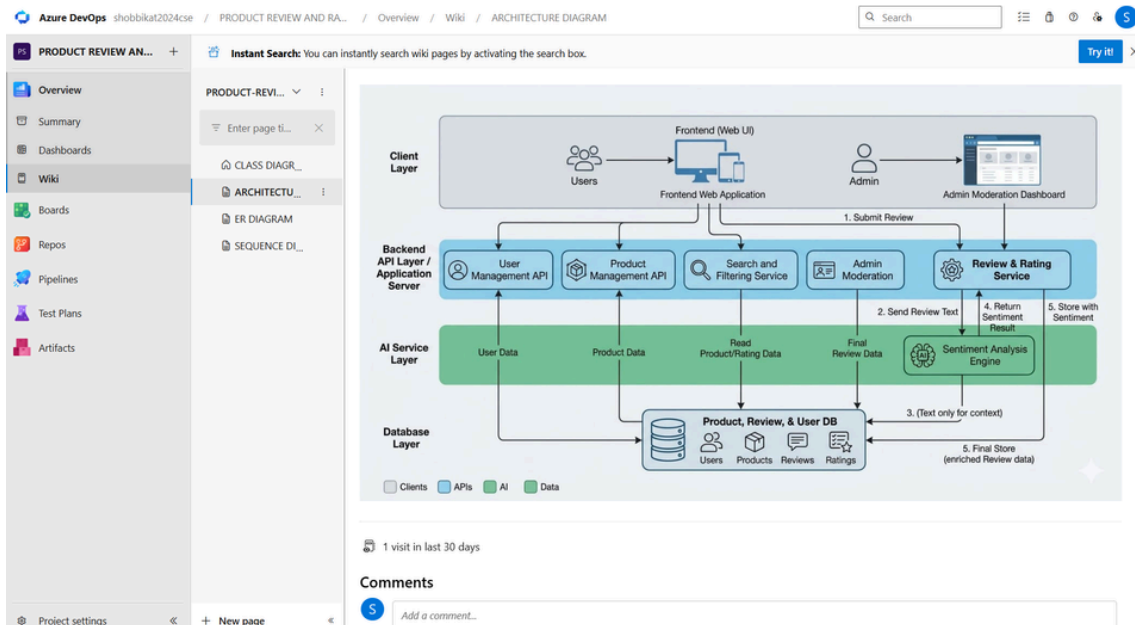
EXP NO: 7

DESIGNING ARCHITECTURAL AND ER DIAGRAMS FOR PROJECT STRUCTURE

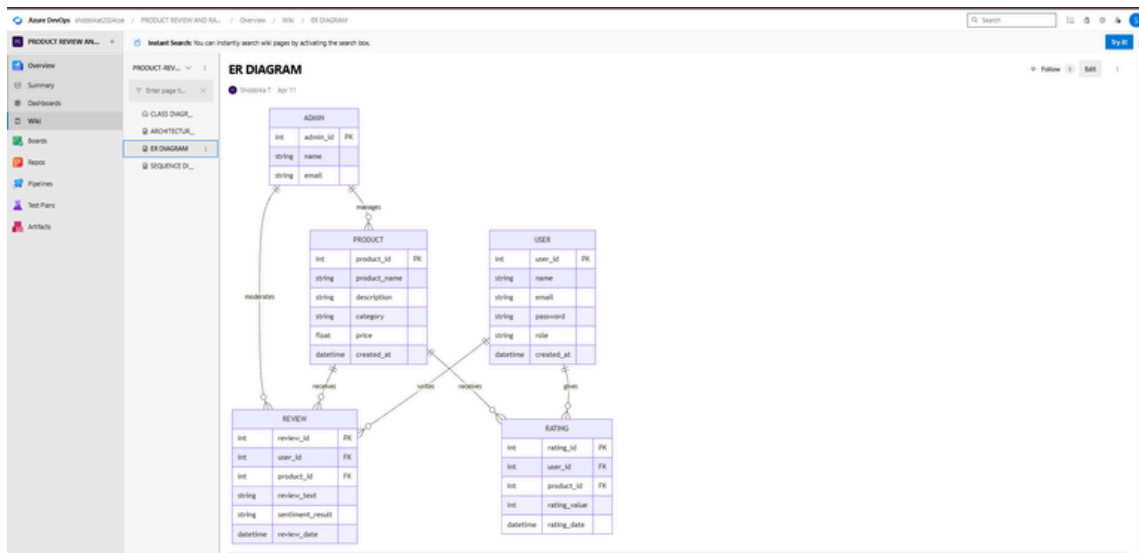
Aim:

To Design an Architectural Diagram and ER Diagram for the given Project.

7A. Architectural Diagram



7B.ER Diagram



Result:

The Architecture Diagram and ER Diagram is designed Successfully for the Product Review and Rating System.

EXP NO: 8	TESTING – TEST PLANS AND TEST CASES
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Aim:

Test Plans and Test Case and write two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Test Planning and Test Case

Test Case Design Procedure

1. Understand Core Features of the Application

- User Registration & Login
- Viewing and Managing Products
- Adding, Editing, and Deleting Products
- Submitting Reviews and Ratings for Products
- Searching, Filtering, and Sorting Products
- Admin Dashboard for Managing Users and Reviews

2. Define User Interactions

- Each test case simulates a real user behavior (User logs in successfully, adds a product, submits a 5-star review, or searches for a product successfully.).

3. Design Happy Path Test Cases

- Focused on validating that all features function as expected under normal conditions.
- Example: User logs in successfully, adds item to playlist, or creates a category-based playlist.

4. Design Error Path Test Cases

- Simulate negative or unexpected scenarios to test robustness and error handling.
- Example: Login fails with invalid credentials, review submission fails with empty fields, product deletion fails for unauthorized users, or search returns no products.

5. Break Down Steps and Expected Results

- Each test case contains step-by-step actions and corresponding expected outcomes.
- Ensures clarity for both testers and automation scripts.

6. Use Clear Naming and IDs

- Test cases are named clearly (e.g., TC01 – Successful User Login, TC02 – Invalid Login Attempt)
- Helps in quick identification and linking to user stories or features.

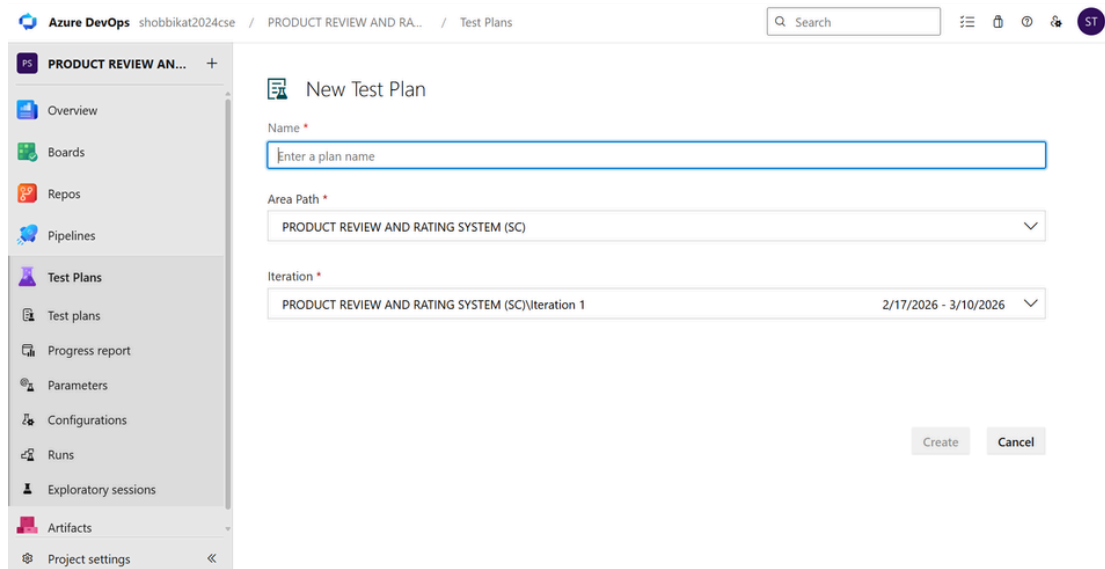
7. Separate Test Suites

- o Grouped test cases based on functionality (e.g., User Management, Product Management, Review & Rating System, Search & Filtering, Admin Features).
- o Improves organization and test execution flow in Azure DevOps.

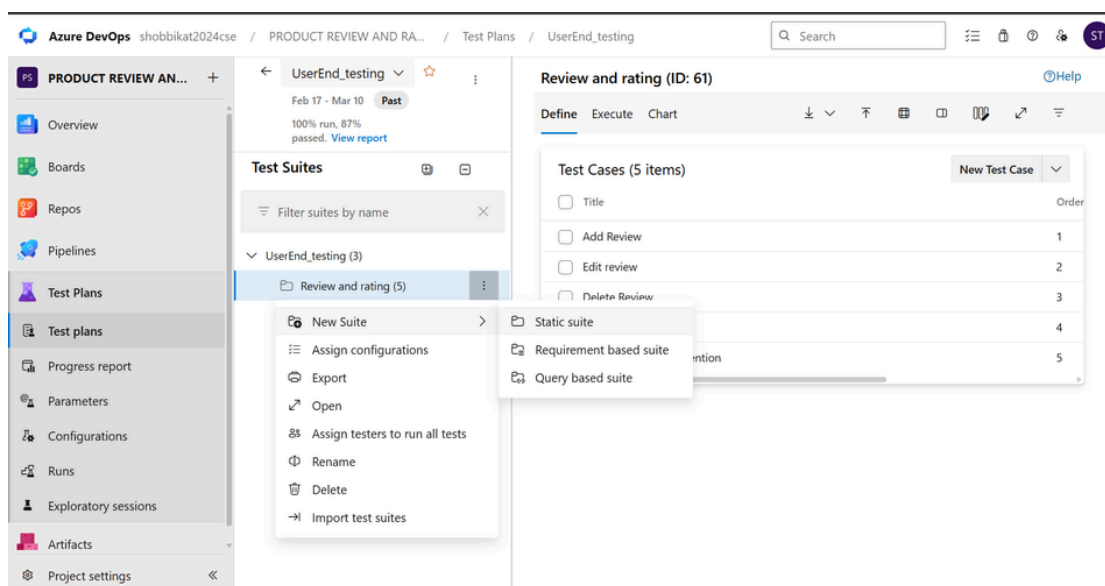
8. Prioritize and Review

- o Core functionalities like login, product management, and review submission are marked high priority.
- o Reviewed for completeness and traceability against feature requirements.

1.New test plan



2.Test suite



3. Test case

Give two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Product Review and Rating System – Test Plans

USER STORIES

- As a user, I want to sign up and log in securely so that I can access the system (ID: 101).
- As a user, I need to view, add, edit and delete products (ID:102)
- As a user, I should be able to submit ratings and reviews for product (ID: 103).
- As a user, I want to search and filter product easily (ID: 104).
- As a admin, I need to manage users, reviews, and products (ID: 105).

Test Suites

Test Suit: TS01 - User Management (ID: 106)

1. TC01 – Successful User Registration

- o **Action:**
 - Goto the Sign-Up page.
 - Enter valid name, email, and password.
 - Click "Register".
- o **Expected Results:**
 - Register form is displayed.
 - Fields accept values without error.
 - Account is created, and the user is redirected to the dashboard.
- o **Type:** Happy Path

2. TC02 – Login with Invalid Password

- o **Action:**
 - Goto the Login page.
 - Enter valid email and password.
 - Click on "Login".
- o **Expected Results:**
 - Login form is displayed.
 - Fields accept data without error.
 - User is logged in and redirected to the dashboard.
- o **Type:** Happy Path

3. TC03 – Sign Up with Existing Email

- o **Action:**
 - Goto the Sign-Up page.
 - Enter a name and an already registered email.

- Click on "Register".
- o **Expected Results:**
 - Fields accept data.
 - Error message "Email already registered" is displayed.
- o **Type:** Error Path

4. TC04 – Login with Wrong Password

- o **Action:**
 - Go to the Login page.
 - Enter valid email and incorrect password.
 - Click on "Login".
- o **Expected Results:**
 - Input is accepted.
 - Error message "Invalid username or password" is shown.
- o **Type:** Error Path

Test Suit: TS02 - Product Management (ID: 107)

1. TC05 – View Products

- o **Action:**
 - Log in successfully.
 - Navigate to "My Products" section.
- o **Expected Results:**
 - All created playlists are displayed clearly.
- o **Type:** Happy Path

2. TC06 – Products Loading Failure

- o **Action:**
 - Disconnect from the internet.
 - Navigate to "My Products".
- o **Expected Results:**
 - Network is offline.
 - Error message "Unable to load products" is shown.
- o **Type:** Error Path

Test Suit: TS03 - Review and Rating System (ID: 108)

1. TC07 – Submit Review Successfully

- o **Action:**
 - Login and Open product details page
 - Enter Review and Click on “Submit Review”
- o **Expected Results:**
 - Review added and Rating updated successfully
- o **Type:** Happy Path

2. TC08 – Submit Empty Review

- o **Action:**
 - Leave review field blank
 - Click “Submit Review”.
- o **Expected Results:**
 - • Error message "Review cannot be empty" is displayed.
- o **Type:** ErrorPath

Test Suit: TS04 - Search & Filtering (ID: 109)

1. TC09 – Search Product by Name

- o **Action:**
 - Navigate to "My Products".
 - Enter product name in search bar.
 - Click "Search".
- o **Expected Results:**
 - Matching products are displayed successfully.
- o **Type:** HappyPath

2. TC10 – Search for Non-Exsisting Product

- o **Action:**
 - Navigate to “My Products”.
 - Enter invalid or unavailable product name.
 - Click "Search".
- o **Expected Results:**
 - Error message "No Products available" is shown.
- o **Type:** Error Path

3. TC11 – Filter Product by Category

- o **Action:**
 - Navigate to products page.
 - Select a category from filter options.
 - Click "Apply Filter".
- o **Expected Results:**
 - Filter works correctly without errors.
- o **Type:** Happy Path

4. TC12 – Filter Returns No Products

- o **Action:**
 - Navigate to products page.
 - Select a category with
 - No available products.
 - Click "Apply Filter".
- o **Expected Results:**
 - Errormessage: "No products found in this category".
- o **Type:** Error Path

Test Suit: TS05 - Admin Features (ID: 110)

1. TC13 – Admin Deletes Inappropriate Review

- o **Action:**
 - Login as admin.
 - Navigate to review management section.
 - Select a review.
 - Click "Delete".
- o **Expected Results:**
 - Deleted review no longer appears.
- o **Type:** HappyPath

2. TC14 – Unauthorized User Accessing Admin Panel

- o **Action:**
 - Login with valid credentials. Click on
 - "Normal user".
 - Select categories.
 - Attempt to open admin dashboard URL.
- o **Expected Results:**
 - Error message: "Access Denied" or "Unauthorized access".
- o **Type:** Error Path

Test Cases

TEST CASE 65*

65 Add Rating

Shobbika T

0 Comments Add Tag

Save and Close

Follow

State: Design

Area: PRODUCT REVIEW AND RATING SYSTEM (SC)

Reason: New

Iteration: PRODUCT REVIEW AND RATING SYSTEM (SC)\Iteration 1

Updated by Shobbika T: Just now

Steps

Summary

Associated Automation

Recent test results

Run manual tests with web runner. [Learn how to get started](#)

Deployment

Development

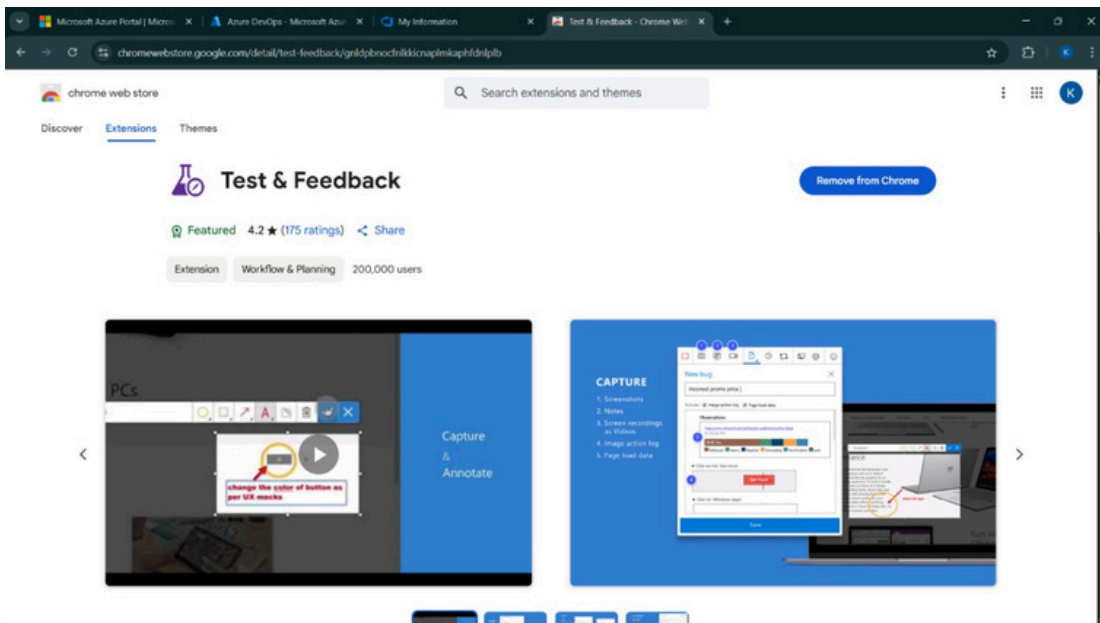
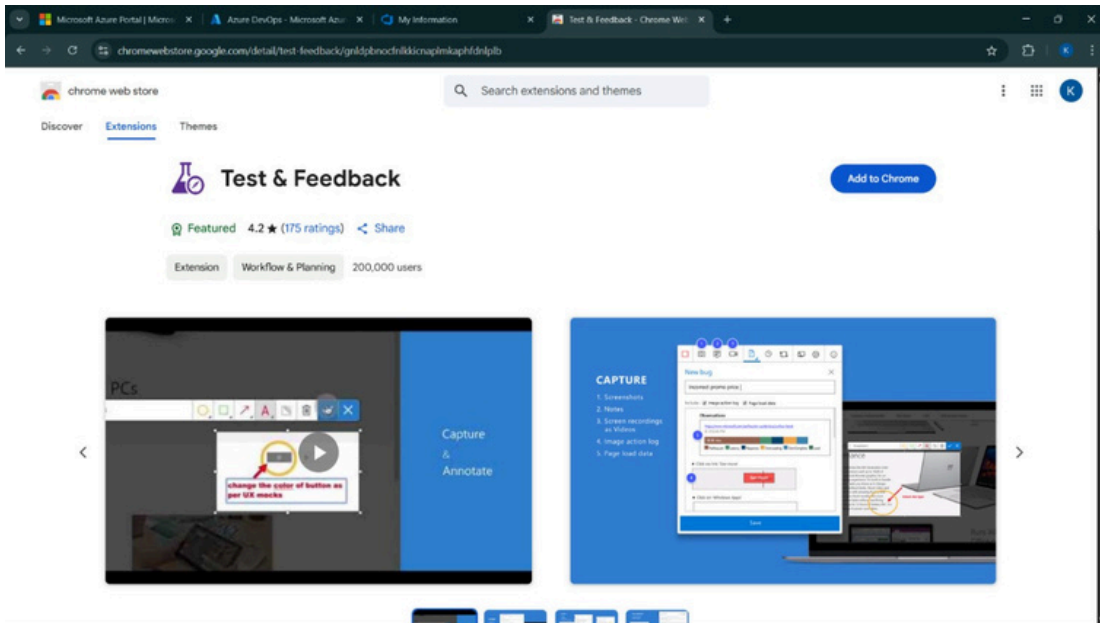
Related Work

Status

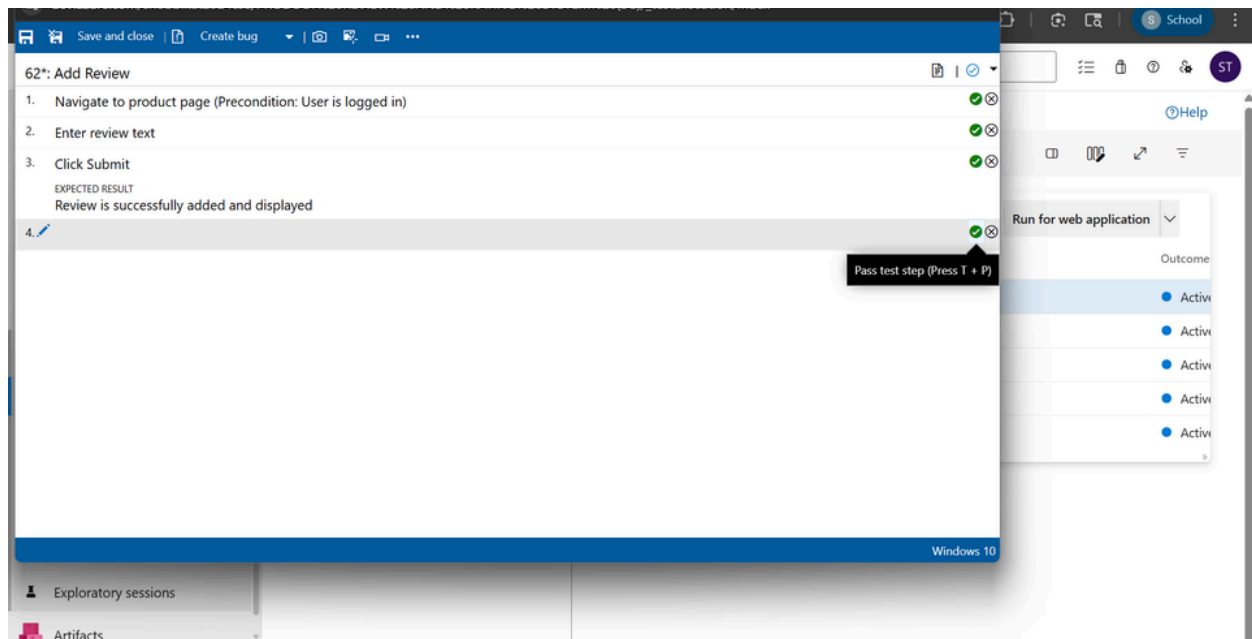
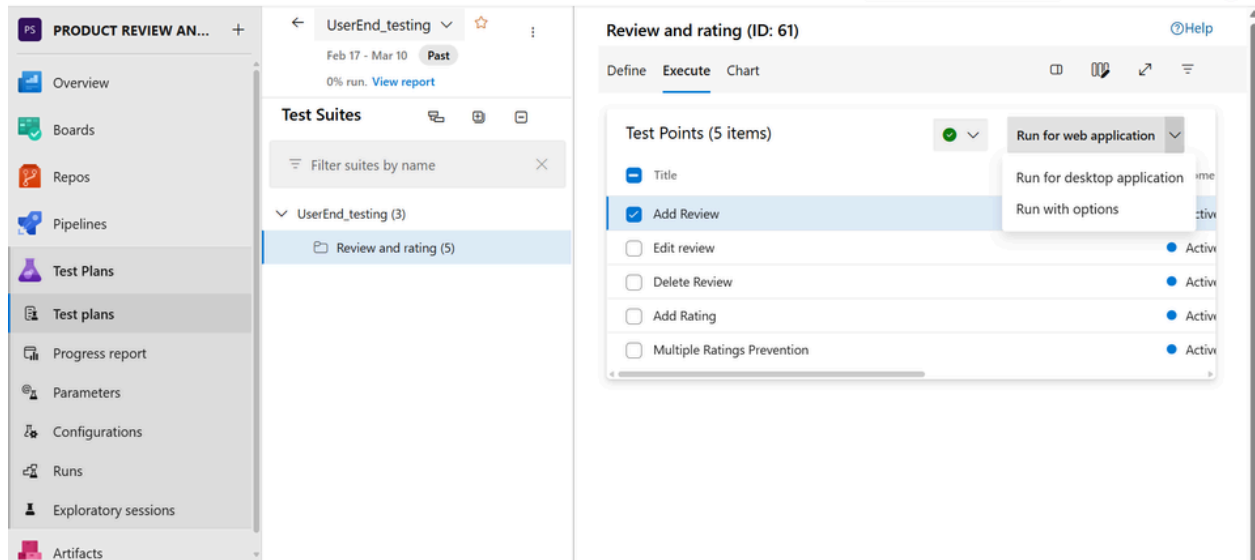
Steps	Action	Expected result
1.	Select rating value (click on star Icons)	
2.	Submit	Rating stored successfully

Click or type here to add a step

4.Installation of test



5. Running the test cases



6. Creating the bug

dev.azure.com/shobbikat2024cse/PRODUCT%20REVIEW%20AND%20RATING%20SYSTEM%20(SC)/_testExecution/Index

Save and close Create bug

71*: add review

1. Navigate to product page (Precondition: User is logged in) ✓ ✗

COMMENT

Add notes or context

2. Enter review text ✓ ✗

COMMENT

Add notes or context

3. Click Submit ✓ ✗

EXPECTED RESULT

Review is successfully added and displayed

Runner - Test Plans - Google Chrome

dev.azure.com/shobbikat2024cse/PRODUCT%20REVIEW%20AND%20RATING%20SYSTEM%20(SC)/_testExecution/Index

NEW BUG *

Unable to submit review. Takes more time when there's poor network

No one selected 0 Comments Add Tag Save and Close

State: New Area: PRODUCT REVIEW AND RATING SYSTEM (SC)

Reason: New Iteration: PRODUCT REVIEW AND RATING SYSTEM (SC)/Iteration 1

Details 2 0

5/11/2026 2:22 PM Bug filed on "add review"

Step no.	Result	Title
1.	Passed	Navigate to product page (Precondition: User is logged in)
2.	Passed	Enter review text
3.	Failed	Click Submit

Expected Result

Story Points

Priority: 2

Severity: 3 - Medium

Activity

PS

PRODUCT REVIEW AN... +

Overview

Boards

Repos

Pipelines

Test Plans

Test plans

Progress report

Parameters

Configurations

Runs

Exploratory sessions

Artifacts

Project settings

Needs investigation

Test run 14 - Review and rating (Manual)

Run summary

Attachments 0

Analytics

0.00% pass rate

1 Failed

Manual test

Test plan 55 - UserEnd_testing

Owned by Shobbika T

Time completed / duration

5/11/2026 2:25 PM

5m 55s

Pipeline run tested

None

Release & release stage

None

Comment

Edit

Filter by Test case ID or Title

Outcome

Configuration

Priority

Owner

Test results

Bug

Link

Group by: None

Outcome	Test case	Run by	Priority	Configuration	Owner	Test suite
Failed	71 add review	Shobbika T	2	Windows 10	Shobbika T	61 Review

8.Test case results

Azure DevOps

Test Plans / Runs / Test run 3 - Review and ra... / 100000 Add Review

Search

PS

PRODUCT REVIEW AN... +

Overview

Boards

Repos

Pipelines

Test Plans

Test plans

Progress report

Parameters

Configurations

Runs

Exploratory sessions

Artifacts

Project settings

Run - 3 Review and ra...

Search by test case ID or title

Filter on

62 Add Review

Windows 10

63 Edit review

Windows 10

Summary

Attachments 0

Analysis

Edit

Analysis owner

Shobbika T

Comment

Linked work items

Add

No work items linked

Test paused

Start time 4/29/2026 4:05 AM

Duration 17s

Show images

9. Test report summary

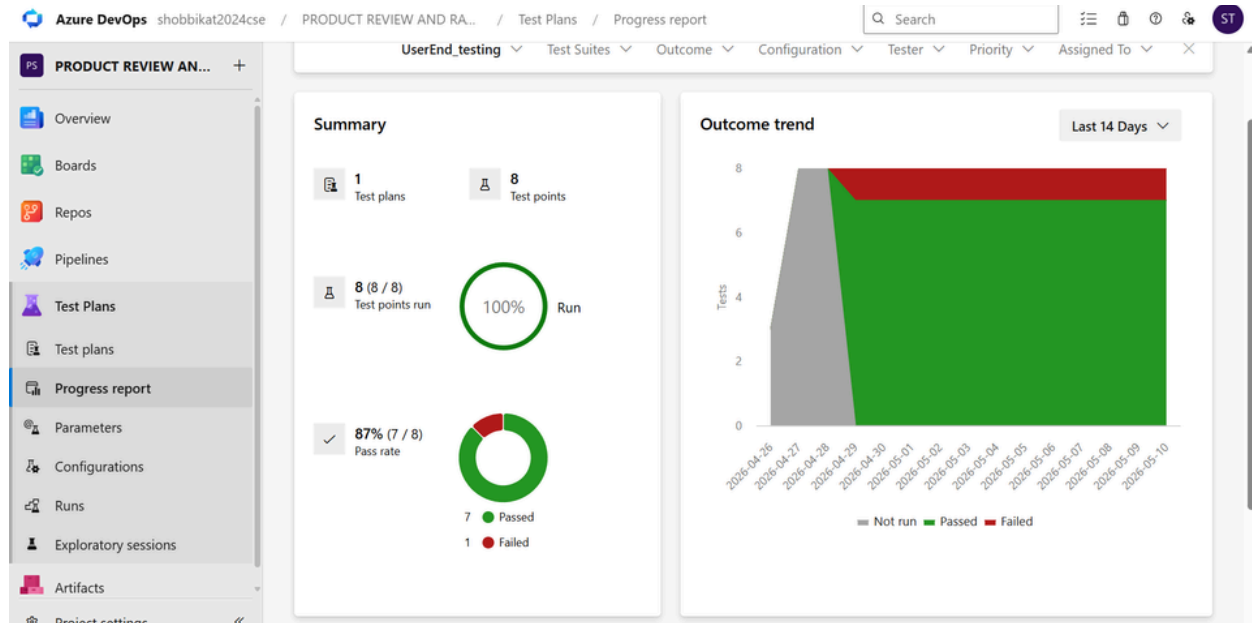
The screenshot shows the Azure DevOps interface for a work item titled "72 Unable to submit review. Takes more time when there's poor network". The work item is in the "New" state. A dropdown menu is open, showing options: New, Active, Resolved, and Closed. The "New" option is selected. The work item details include a description, a reason, and a repro step table.

Step no.	Result	Title
1.	Passed	Navigate to product page (Precondition: User is logged in)
2.	Passed	Enter review text
3.	Failed	Click Submit

The "Expected Result" is also visible. The right sidebar shows the "Planning" and "Deployment" sections, including a "Resolved Reason" and a "Development" section with a link to "Link a GitHub commit".

- Assigning bug to the developer and changing state

10. Progress report



11.Changing the test template

The screenshot shows the Azure DevOps interface for managing process templates. The left sidebar is titled 'Organization Settings' and includes sections for General, Security, Boards, and Pipelines. The 'Process' option under Boards is selected. The main content area, titled 'All processes', displays a table of process templates. The 'Agile' template is expanded, showing its description and team project count.

Name	Description	Team projects
Basic	This template is flexible for any process and great for teams getting started with Azure DevOps.	0
Agile	This template is flexible and will work great for most teams using Agile planning methods, including those practicing Scrum, Kanban, and Hybrid.	0
Scrum	This template is for teams who follow the Scrum framework.	0
CMMI	This template is for more formal projects requiring a framework for process improvement and an auditable record.	0

Microsoft Azure Portal | M x Azure DevOps - Microsoft x My Information x Test Plan 84 Music Playlist x Settings - Process (231801095) x Settings - Overview (231801095) x + -

dev.azure.com/231801095/_settings/process

Azure DevOps 231801095 / Settings / Process

Organization Settings 231801095

Search Settings

General

- Overview
- Projects
- Users
- Billing
- Global notifications
- Usage
- Extensions
- Microsoft Entra

Security

- Security overview
- Policies
- Permissions

Boards

- Process

Pipelines

- Agent pools
- Settings
- Deployment pools

All processes

Processes Fields

Name	Description	Team projects
Basic	This template is flexible for any process and great for teams getting started with Azure DevOps.	0
Agile	This template is flexible and will work great for most teams using Agile planning methods, including those practicing Scrum.	0
Scrum	This template is for teams who follow the Scrum framework.	0
CMMI	This template is for more formal projects requiring a framework for process improvement and an auditable record of process.	0

Microsoft Azure Portal | M x Azure DevOps - Microsoft x My Information x Test Plan 84 Music Playlist x Settings - Process (231801095) x Settings - Overview (231801095) x + -

dev.azure.com/231801095/_settings/process

Azure DevOps 231801095 / Settings / Process

Organization Settings 231801095

Search Settings

General

- Overview
- Projects
- Users
- Billing
- Global notifications
- Usage
- Extensions
- Microsoft Entra

Security

- Security overview
- Policies
- Permissions

Boards

- Process

Pipelines

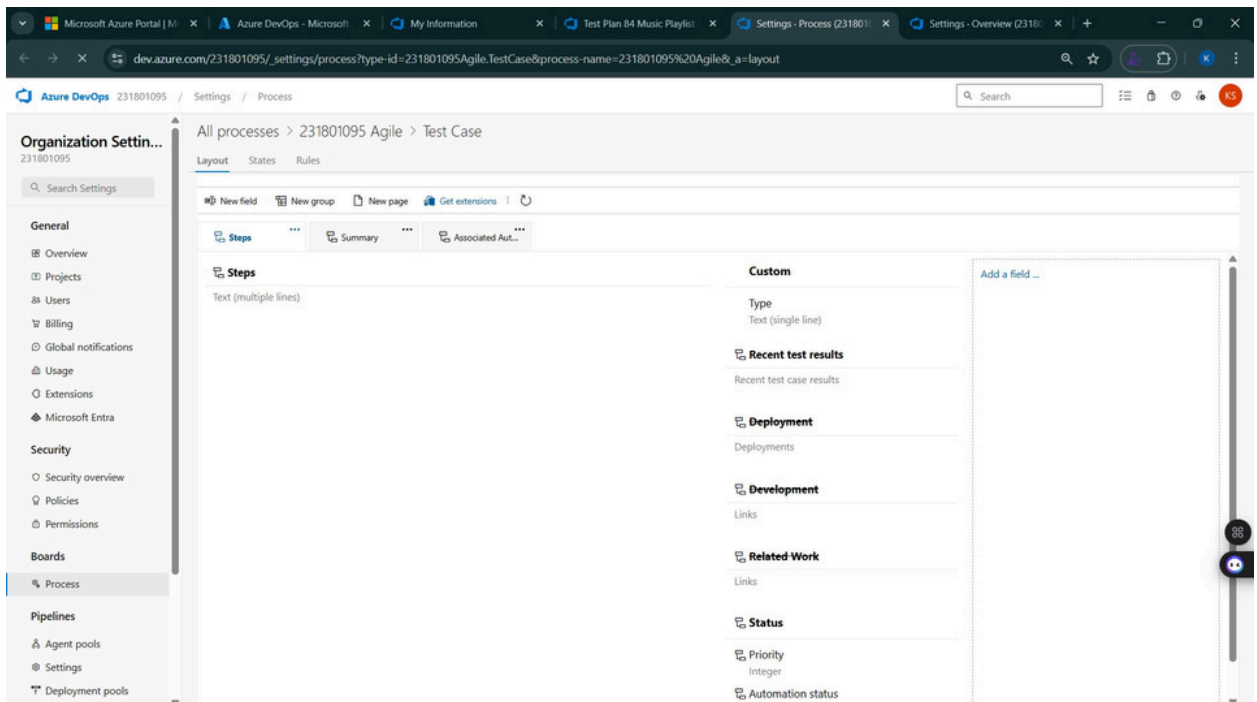
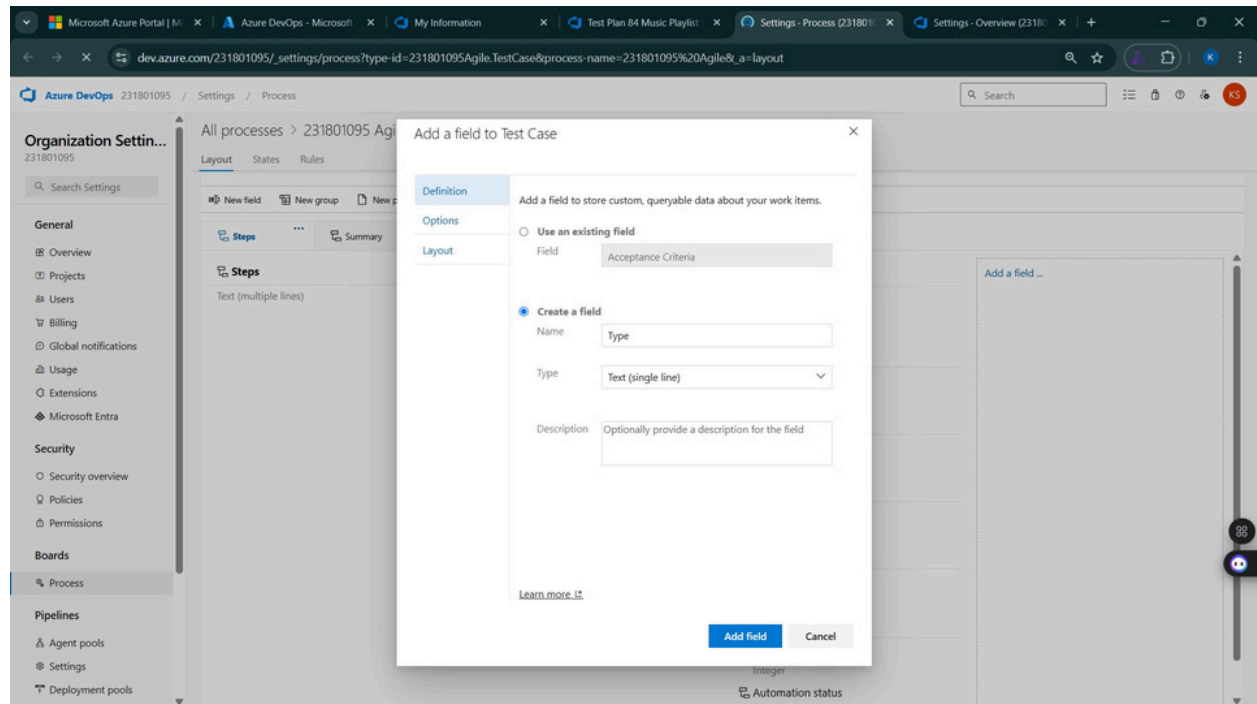
- Agent pools
- Settings
- Deployment pools

All processes

Processes Fields

Name	Description	Team projects
Basic	This template is flexible for any process and great for teams getting started with Azure DevOps.	0
Agile	This template is flexible and will work great for most teams using Agile planning methods, including those practicing Scrum.	0
231801095 Agile (default)		1
Agile Plus		0
Scrum	This template is for teams who follow the Scrum framework.	0
CMMI	This template is for more formal projects requiring a framework for process improvement and an auditable record of process.	0

12.View the new test case template



Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path

EXP NO: 9	LOAD TESTING AND PIPELINES
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Aim:

To create an Azure Load Testing resource and run a load test to evaluate the performance of a target endpoint and to create and demonstrate an Azure DevOps pipeline for automating application builds, tests, and deployment.

Load Testing

Azure Load Testing:

Azure Load Testing allows you to simulate high traffic and stress tests for your web applications and APIs to understand how they perform under load. It helps identify performance bottlenecks, scalability issues, and optimize resource usage before deployment.

Steps to Create an Azure Load Testing Resource:

Before you run your first test, you need to create the Azure Load Testing resource:

1. Sign in to Azure Portal
Go to <https://portal.azure.com> and log in.
2. Create the Resource
 - o Go to *Create a resource* → Search for “Azure Load Testing”.
 - o Select Azure Load Testing and click Create.
3. Fill in the Configuration Details
 - o *Subscription*: Choose your Azure subscription.
 - o *Resource Group*: Create new or select an existing one.
 - o *Name*: Provide a unique name (no special characters).
 - o *Location*: Choose the region for hosting the resource.
4. (Optional) Configure tags for categorization and billing.
5. Click Review + Create, then Create.
6. Once deployment is complete, click Go to resource.

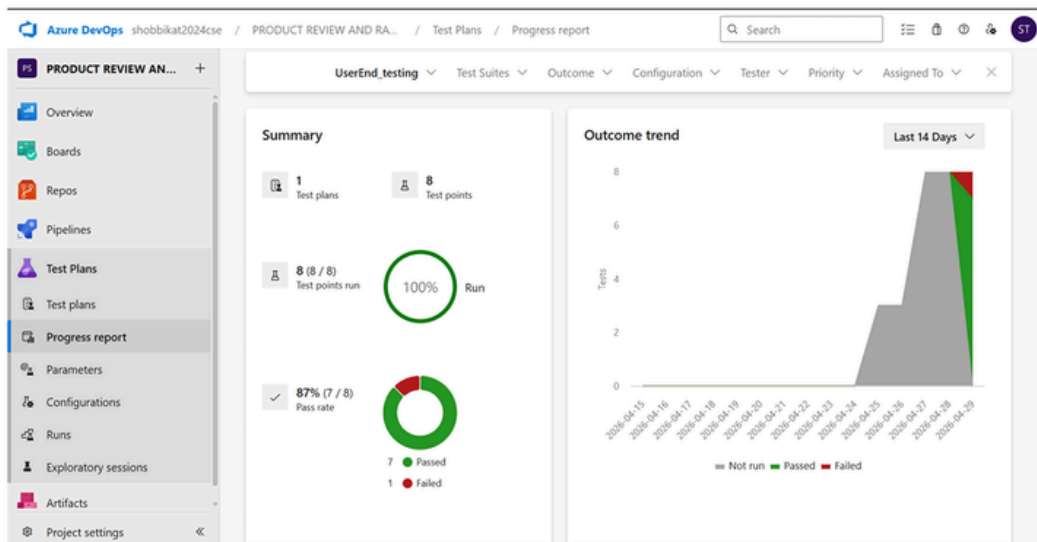
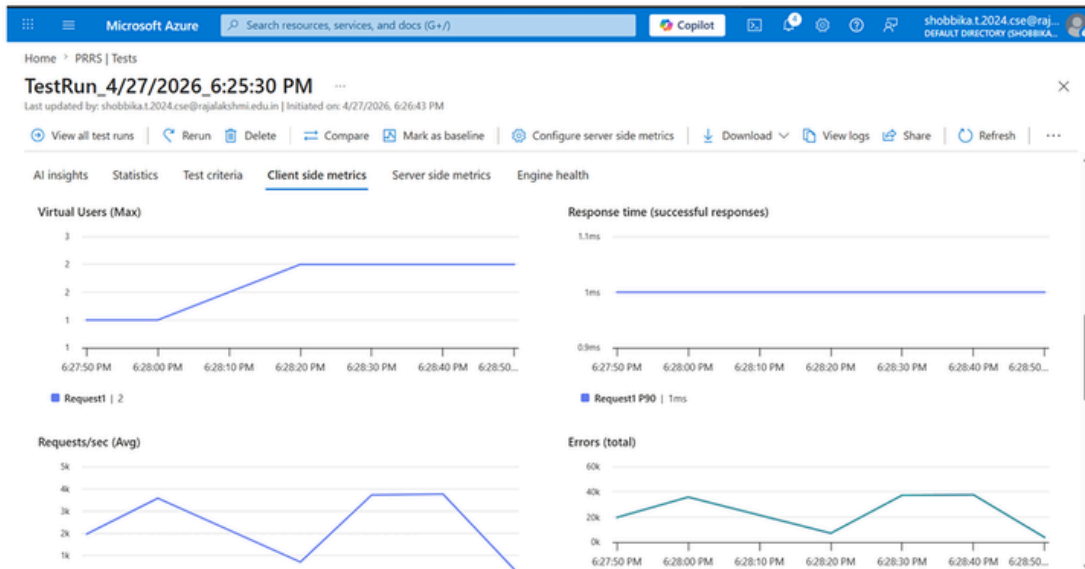
Steps to Create and Run a Load Test:

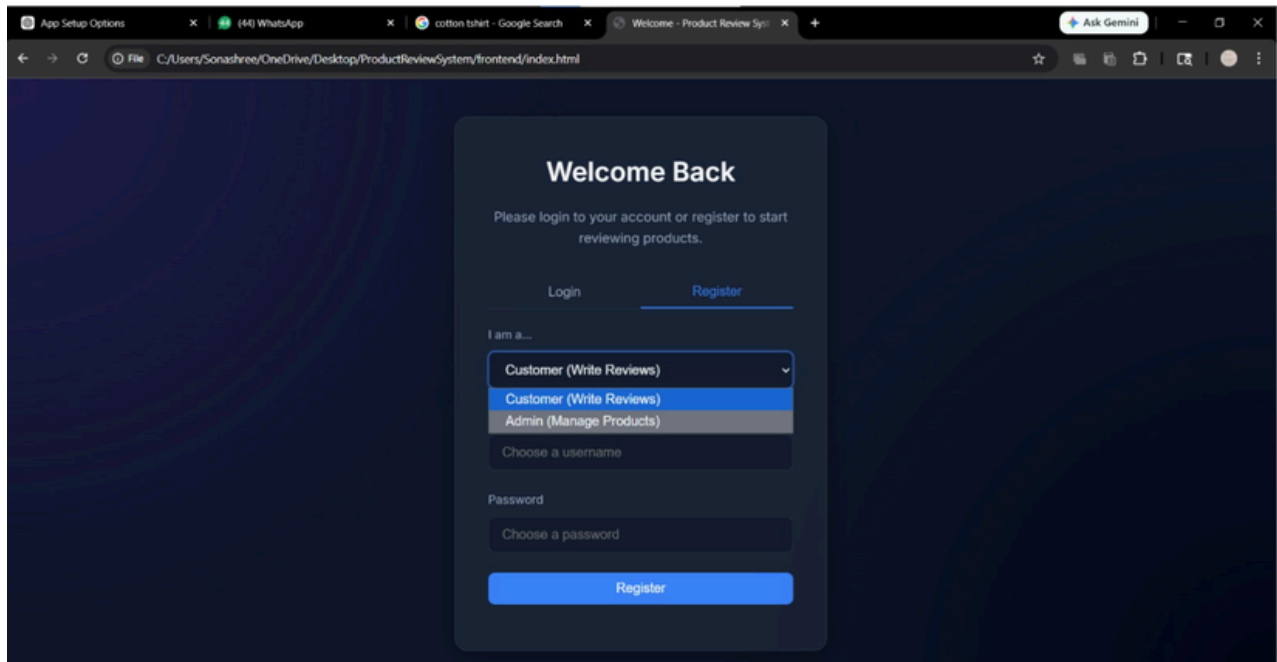
Once your resource is ready:

1. Go to your Azure Load Testing resource and click Add HTTP requests > Create.
2. Basics Tab
 - o *Test Name*: Provide a unique name.
 - o *Description*: (Optional) Add test purpose.
 - o *Run After Creation*: Keep checked.
3. Load Settings
 - o *Test URL*: Enter the target endpoint (e.g., <https://yourapi.com/products>).

- Click Review + Create → Create to start the test.

Load Testing





Pipelines

Description:

This experiment demonstrates how to connect a GitHub-hosted Flask-based music recommendation project with Azure DevOps. The pipeline will automatically install dependencies, run basic tests, and publish artifacts. This ensures that every commit triggers checks for reliability and smooth deployment.

Steps:

1. Connect GitHub to Azure DevOps:
 - o In Azure DevOps, create a new project.
 - o Create a pipeline and select GitHub as the source.
 - o Authorize access to your GitHub repository, ensuring that Azure DevOps can pull the repository for your pipeline.
2. Create azure-pipelines.yml in Your Repo Root:
 - o In your GitHub repository, create a new file called azure-pipelines.yml in the root directory.
 - o Add the following basic pipeline configuration for Python and Flask:

yml Code

trigger:

-main # Trigger pipeline when changes are pushed to the main branch

pool:

vmImage: ubuntu-latest # Use a hosted Ubuntu agent

steps:

#Step 1: Checkout the code from GitHub

-checkout: self

#Step 2: Set up Python environment

-task: UsePythonVersion@0

inputs:

versionSpec: '3.x' # Use the latest Python 3.x version

displayName: "Set up Python"

#Step 3: Install dependencies from the correct path

-script: |

python -m pip install --upgrade pip

pip install -r project/requirements.txt # Adjusted path to requirements.txt

displayName: "Install dependencies"

#Step 4: Run a simple Python script to check the environment

-script: |

python -c "print('Rate and review your favourite products.')"

displayName: "Run a Python script"

3. Pipeline Tasks Include:

- o Setting up the Python environment using the UsePythonVersion task.
- o Installing project dependencies from project/requirements.txt. Make sure the path to requirements.txt is correct (it is located under the project folder).
- o Running a simple Python script to verify that Python is set up correctly and the pipeline works.

4. Run and Monitor Pipeline:

- o Commit changes to the main branch of your repository to trigger the pipeline in Azure DevOps.
- o Monitor the logs in the Azure DevOps portal to view logs, errors, or success messages and ensure everything runs smoothly.

Pipeline

The screenshot displays the Azure DevOps web interface for a project named 'PRODUCT REVIEW AND RATING SYSTEM'. The left sidebar contains navigation links for Overview, Boards, Repos, Pipelines, Environments, Library, Test Plans, and Artifacts. The main area shows the 'Review your pipeline YAML' step, which is part of a new pipeline. The pipeline YAML code is displayed in a text editor, showing the trigger, variables, and the main task. The code is as follows:

```
1 # Python to Linux Web App on Azure
2 # Build your Python project and deploy it to Azure as a Linux Web App.
3 # Change python version to one thats appropriate for your application.
4 # https://docs.microsoft.com/azure/devops/pipelines/languages/python
5
6 trigger:
7   - main
8
9 variables:
10  # Azure Resource Manager connection created during pipeline creation
11  azureServiceConnectionId: 'f8a56bfe-502c-484a-929f-cda157746e43'
12
13  # Web app name
14  webAppName: 'ProductReviewAndRatingSystem'
15
16  # Agent VM image name
17  vmImageName: 'ubuntu-latest'
18
19  # Environment name
20  environmentName: 'ProductReviewAndRatingSystem'
```

Result:

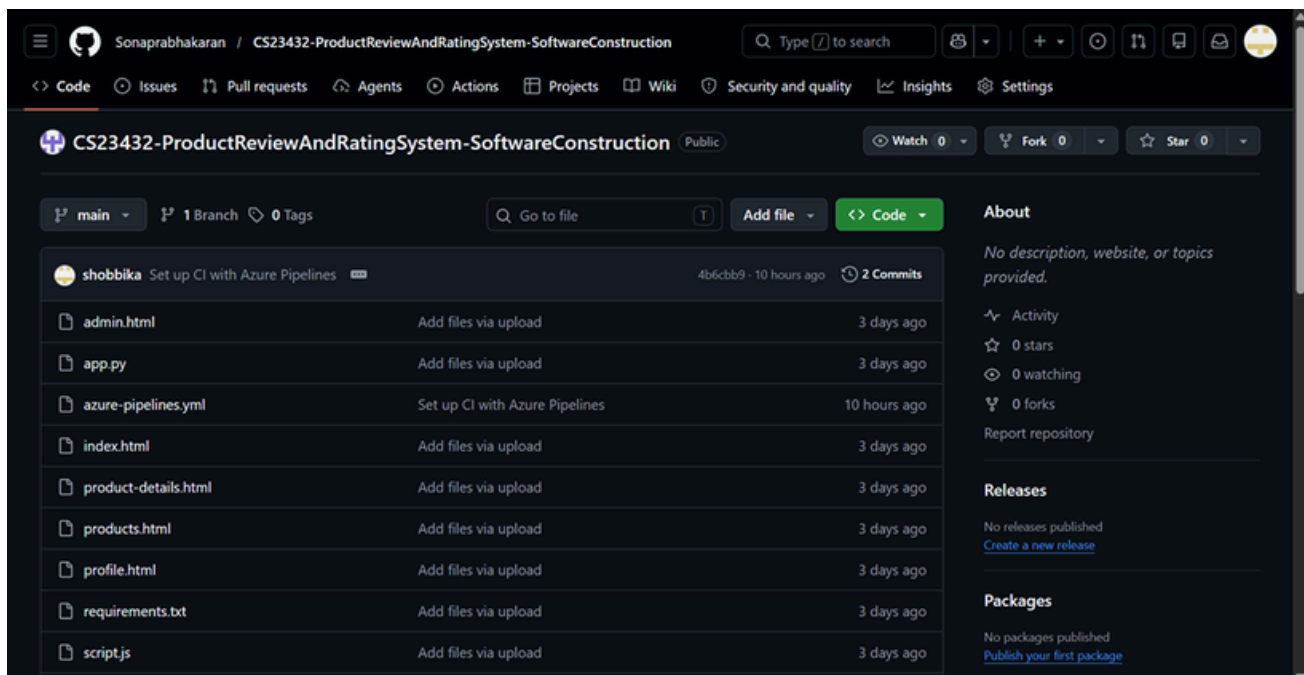
Successfully created the Azure Load Testing resource and executed a load test to assess the performance of the specified endpoint and also demonstrated pipelines in azure devops.

EXP NO: 10	GITHUB: PROJECT STRUCTURE & NAMING CONVENTIONS
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Aim:

To provide a clear and organized view of the project's folder structure and file naming conventions, helping contributors and users easily understand, navigate, and extend the Product review and rating system project.

GitHub Project Structure



Result:

The GitHub repository clearly displays the organized project structure and consistent naming conventions, making it easy for users and contributors to understand and navigate the codebase.

